



Reproducible research in sport and exercise psychology: The role of sample sizes



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ARTICLE INFO

Article history:

Received 27 July 2015

Received in revised form

4 November 2015

Accepted 20 November 2015

Available online 2 December 2015

Keywords:

Replicability

Power

False positive

Effect size

Research methods

ABSTRACT

Objectives: We aim to introduce the discussion on the crisis of confidence to sport and exercise psychology. We focus on an important aspect of this debate, the impact of sample sizes, by assessing sample sizes within sport and exercise psychology. Researchers have argued that publications in psychological research contain numerous false-positive findings and inflated effect sizes due to small sample sizes.

Method: We analyse the four leading journals in sport and exercise psychology regarding sample sizes of all quantitative studies published in these journals between 2009 and 2013. Subsequently, we conduct power analyses.

Results: A substantial proportion of published studies does not have sufficient power to detect effect sizes typical for psychological research. Sample sizes and power vary between research designs. Although many correlational studies have adequate sample sizes, experimental studies are often underpowered to detect small-to-medium effects.

Conclusions: As sample sizes are small, research in sport and exercise psychology may suffer from false-positive results and inflated effect sizes, while at the same time failing to detect meaningful small effects. Larger sample sizes are warranted, particularly in experimental studies.

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“At its core, this crisis is about the justification of knowledge. Researchers in the social and behavioral social sciences make claims that are simply unsupported by the methods they use.”

With this quote, Hoekstra, Morey, Rouder, and Wagenmakers (2014, p. 1162) refer to the so-called crisis of confidence that has been an alarming topic in psychological research during the past years. While the crisis of confidence has led to numerous publications (e.g., Fraley & Vazire, 2014; Open Science Collaboration, 2012, *in press*), several special issues (e.g., Nosek & Lakens, 2014; Pashler & Wagenmakers, 2012; Spellman, 2012) and even the emergence of new organizations or projects dedicated to improving the quality of psychological research (Open Science Collaboration, 2012; *in press*), the field of sport and exercise psychology has been relatively unaffected by this movement. This does not mean that scholarly contributions calling for high standard research methodology have been absent within this field: see for example Myers, Ahn, and Jin (2011) on a special instance of estimating sample sizes and power, or Zhu (2012) on Null Hypothesis

Significance Testing, or reliability (Zhu, 2013). However, an alarming series of failed replications (Open Science Collaboration, 2015) has cast justified, new doubt on common research and publishing practices. In a first step of remedying the main problems causing the failed replications, the field of psychology has provided evidence for some of the most important shortcomings within psychology. Following this approach the present manuscript has the specific aim of introducing this discussion to sport and exercise psychology, and more importantly of reviewing the state of the field regarding a crucial aspect of the discussion, namely the adequacy of sample sizes.

1. The current debate: a crisis of confidence

The so-called *crisis of confidence*, sometimes also called *replication crisis* centers around the observation that it has been surprisingly hard to replicate well-known psychological results (e.g., Ebersole et al., 2015; Nosek & Lakens, 2014; Open Science Collaboration, 2015). Furthermore, it has been noted that replications have been underappreciated by the scientific community and publication outlets and therefore replication attempts are hard to find in the psychological literature (Makel, Plucker, &

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Hegarty, 2012; Open Science Collaboration, 2012). At the same time, researchers have cast doubt on the quality of the research methods psychologists commonly use (Simmons, Nelson, & Simonsohn, 2011; Wagenmakers, Wetzels, Borsboom, & van der Maas, 2011). For example, several papers have compellingly argued that psychological research suffers from an inflated rate of false-positives or Type 1 errors, and they have identified factors contributing to the inflation of false-positives (Simmons et al., 2011). A false-positive means that researchers report finding an effect that does not exist in the real world. In other words, researchers reject the null hypothesis although it would have been true (Simmons et al., 2011). Although most researchers may be aware of some risk of reporting a false-positive, the problem is that many researchers do not seem to be aware of the magnitude of this risk and of what factors contribute to it. The actual risk of reporting a false-positive depends on several factors, and given certain constellations of factors it may well exceed 50%, as Simmons and colleagues convincingly showed (2011).

... we show that despite empirical psychologists' nominal endorsement of a low rate of false-positive findings ($\leq .05$), flexibility in data collection, analysis, and reporting dramatically increases actual false-positive rates. In many cases, a researcher is more likely to falsely find evidence that an effect exists than to correctly find evidence that it does not ... (Simmons et al., 2011, p. 1359)

This number is neither a new one nor confined to psychological research: As early as in 2005, Ioannidis warned that in biomedical research about 50% of all reported research findings may be wrong (2005). But why is that? The inflation of false-positive rates is usually explained by the interplay of several factors (Ioannidis, 2005; Simmons et al., 2011). The most prominent among them are researcher degrees of freedom, the file-drawer effect, and sample sizes.

Researcher degrees of freedom refer to choices researchers can make when collecting, analyzing, and reporting their data (Simmons et al., 2011). For example, researchers may sequentially increase their sample size, choose between several covariates or between several dependent variables, or they may selectively drop experimental conditions. When researchers employ any of these behaviours or combinations of them, and the null hypothesis is actually true, then the probability that their study will yield a false-positive result increases. In turn, when significant results have a higher likelihood of being published, the proportion of false-positive results in relation to correct-positive results gets higher (Simmons et al., 2011). When systematic replication attempts are scarce, these factors lead to an ever increasing proportion of false positive results, that may go undetected for a long time. Whereas the contribution of researcher degrees of freedom and selective publishing of significant results is intuitively appealing, the role of sample sizes is far less intuitive to understand. This is despite the fact that sample sizes may play the most important role, both in understanding the crisis of confidence and in finding remedies against this crisis.

2. Adequate sample sizes

Traditionally, sample sizes have been discussed within the power framework (Cohen, 1962, 1992; Sedlmeier & Gigerenzer, 1989). Power is typically defined in terms of Null Hypothesis Significance Testing (NHST). In NHST, power is defined as the

probability of getting a significant result in a study, given the null hypothesis is wrong. In other words, the power of a test is the probability of correctly rejecting the null hypothesis (Button et al., 2013; Gelman & Carlin, 2014). Within this framework, two kinds of errors can be made. A Type 1 error means rejecting the null hypothesis although it is true, whereas a Type 2 error means accepting the null hypothesis although it is false. A Type 1 error is sometimes also called a false-positive and a Type 2 error a miss (Fraleigh & Vazire, 2014).

Some researchers deliberately refrain from referring to the concepts power, Type 1 error, and Type 2 error because these are only relevant from a NHST point of view which has received increasing criticism lately (Cumming, 2012; 2014; Gelman & Carlin, 2014). Alternatively, Gelman and Carlin (2014) suggest using the terms Type M error and Type S error. A Type M error is an error of magnitude that means it refers to the degree to which the estimate of an effect in a study deviates from the real effect. A Type S error is a sign error that means it occurs when the estimate of an effect in a study has the wrong sign. The main difference between the former and the latter concepts is that the former (i.e., Type 1 error and Type 2 error) refer to a binary concept, i.e. a hypothesis is either rejected or it is not. The latter (i.e., Type S error and Type M error) assume that we want to estimate an effect and that we can quantify the degree to which the estimate is off. It is important to note that the relevance of sample sizes is not confined to studies using NHST. In recent years, opposition to NHST (see also Zhu, 2012 within the field of sport science) has increased, and many researchers strongly suggest abolishing NHST for most study purposes (Cohen, 1994; Cumming, 2012; Wagenmakers, 2007). The journal *Basic and Applied Social Psychology* has even banned NHST in research articles (Trafimow & Marks, 2015). Suggested alternatives are the estimation of confidence intervals around effect sizes (Cumming, 2012; 2014) and Bayesian methods (Lee & Wagenmakers, 2005; Wagenmakers, 2007; Wetzels et al., 2011). Both parameter estimation and Bayesian methods require adequate sample sizes as well. Therefore, the following discussion is not limited to studies using NHST.

2.1. The role of sample sizes for power

There are several reasons why sample sizes matter. The first reason is that power depends on sample size. That means, the larger a sample, the higher is the probability to find an effect, given it really exists (Cohen, 1962; 1992). This observation is not new, and researchers have been aware of it for a long time (e.g., Cohen, 1962). Nevertheless, psychological studies still often suffer from low power (Bakker, van Dijk, & Wicherts, 2012; Button et al., 2013; Fraleigh & Vazire, 2014; Sedlmeier & Gigerenzer, 1989). For example, Bakker and colleagues estimate the typical power in psychological research to be around .35 for studies that compare two independent samples when assuming an effect size of $d = .50$ (2012). Button et al. estimate the median power in neuroscience to be around .21 (2013).

Whereas most researchers are aware that smaller samples have a higher likelihood of producing a miss (i.e. of not yielding a significant test although the effect exists), many are not aware of the seemingly paradoxical fact that smaller samples also have a higher likelihood of producing a false-positive (i.e. of yielding a significant test although the effect does not exist). Low power "... negatively affects the likelihood that a nominally statistically significant finding actually reflects a true effect" (Button et al., 2013). This is due to the fact that the positive predictive value PPV (the post-study probability that a claimed effect is true) depends on the

power of a study.¹

2.2. The role of sample sizes for the precise estimation of effect sizes

Furthermore, smaller samples have a higher likelihood of producing a Type M error (i.e. of yielding an effect size which is considerably larger than the true effect). Effect sizes in small sample studies are biased upwards, because smaller samples are more likely to be affected by chance variation in the sampled data. The key to understanding this phenomenon is to understand that data vary randomly and therefore, effects fluctuate. The smaller a sample is, the higher the influence of a few strong variations. At the same time, when samples are small, only large effects pass the statistical significance filter. When effects need to pass a certain threshold in order to be regarded as a discovery, studies with accidentally larger than true effects get published, whereas studies with accidentally smaller than true effects are discarded (Button et al., 2013; Ioannidis, 2005). This phenomenon has been labeled “the winner’s curse” (Button et al., 2013): The researcher who is the first to claim a discovery is cursed by having a high likelihood of reporting an inflated effect size. The influence of small-sample variation is particularly troublesome when researchers subsequently sample more participants until a test produces a significant result and then stop sampling (Simmons et al., 2011). This approach is bound to produce false positive results, as it takes advantage of the first time that (random) variation makes the p-value drop below the significance threshold, although it might well exceed it again. However, researchers and reviewers sometimes think that it is particularly impressive to find an effect with a small sample. Simmons and colleagues (2011, p. 1362) refer to this misconception as „... the often-espoused yet erroneous intuition that if an effect is significant with a small sample size then it would necessarily be significant with a larger one“. In addition to the above points, small sample studies are also more likely to be affected by biasing influences such as researcher degrees of freedom (Button et al., 2013). As smaller samples have more variation, there is more variation that can be exploited by researcher degrees of freedom.

2.3. The role of sample sizes for replicability

Finally, small samples in a field make replication attempts particularly difficult. If an effect exists, the probability of finding it in two unrelated studies (i.e. the probability of replicating an existing effect) is the product of the power of both studies. When samples are small, a failure to replicate (i.e. to find the same effect in both studies) is ambiguous: It may either mean that the effect really does not exist or that the study did not have enough power to detect it. Additionally, the smaller samples in a field are, the more effect sizes vary, which makes it likely that effects from different studies differ.² Therefore, a field with underpowered

studies “... is likely to be full of “failures to replicate” and “inconsistent findings”” (Fraleley & Vazire, 2014, p. 4). The role of sample sizes for replications is indeed so crucial that Fraley and Vazire conclude:

... It is possible that the so-called *replication crisis* or *crisis of confidence* in psychology is an artifact of a long history of researchers using under-powered designs – a state of affairs that is guaranteed to reduce the odds that published findings can be replicated. (2014, p. 4)

3. The N-Pact factor

Therefore, sample sizes are important for quality research in multiple ways. First, adequately powered studies are a prerequisite for finding an effect, given it really exists. Second, adequately powered studies protect from an inflated rate of false positive findings and an inflated rate of Type S and Type M errors. Third, adequately powered studies are a prerequisite for replication. Recently, Fraley and Vazire introduced the N-Pact Factor (2014). They calculated the N-Pact Factor (NF) as the median sample size of the studies a journal publishes per year. The larger the median sample, the higher the NF. Fraley and Vazire analysed sample sizes from six journals in social and personality psychology over a five-year period. First, Fraley and Vazire find that sample sizes in their sample were generally rather small. They report that the typical study in their sample did not have sufficient power (i.e. .80) to detect an effect size of $r = .20$.³ Second, they find larger samples in personality psychology than in social psychology. Third, they report a negative correlation between a journal’s impact factor and its N-Pact Factor. This association might be caused by the fact that counter intuitive results are cited more often, but at the same time it is more likely to obtain a counter intuitive result when using small samples.

Fraley and Vazire intend the NF to be a new metric for ranking and comparing journals based on the statistical power of studies they have published. In this vain, they suggest the NF might be used for authors choosing where to submit their work, for hiring committees comparing the publication records of applicants and, maybe most importantly, as an incentive for journals in order to improve their statistical quality by increasing sample sizes. The question whether the N-Pact Factor will indeed be adopted and routinely used by the scientific community, and whether this will really be beneficial for psychological science can only be resolved by a thorough discussion within the psychological community. Trying to answer this question is therefore beyond the scope of this paper. However, we agree with Fraley and Vazire that sufficiently large samples are of utmost importance and that many fields of psychological research need larger samples in order to obtain credible and reproducible results. Thus far, it is unclear whether the finding that the “typical study” in the field of personality and social psychology did not have adequate power to detect small-to-medium effects is a peculiarity of this field of psychological science or whether this might be true for other branches, too. Within the field of sport and exercise psychology, there is currently no information on whether the size of samples in this field can typically be considered adequate or not. Therefore, we do not know whether existing results can be considered credible and

¹ For more in-depth illustrations why smaller samples produce more false positives see Ioannidis (2005), Button et al. (2013) or Wagenmakers and colleagues (in press). In short, the PPV is $(1 - \beta) \times R / (1 - \beta) \times R + \alpha$. R is the pre-study probability that an effect exists. It follows that for a given R and given α the lower the power the lower the PPV (Button et al., 2013). Put differently (Ioannidis, 2005), a finding is more likely to be true than false when $(1 - \beta) \times R > \alpha$. The relationship between power and PPV becomes more evident when one does not think about the power and PPV of a single study, but about a collection of studies in a certain field (see Button et al., 2013). In this field, we assume a certain R, a certain convention on α and an average power $(1 - \beta)$.

² The winner’s curse as well has profound effects on replication attempts (Button et al., 2013): When researchers use the first published effect size as an estimate for planning of their sample size, and when this effect size is inflated, then they cannot find it, because their sample is too small. In turn, the effect is said to be unreplicable, whereas in truth it may exist, although at a much smaller level than anticipated.

³ Obviously, all published power estimates depend on assumptions of effect sizes. For example, whereas Fraley and Vazire (2014) report that the typical study in their sample did not have sufficient power (i.e. .80) to detect an effect size of $r = .20$, for five out of six journals in their sample power exceeded .80 for effect sizes of $r = .30$.

reproducible and we cannot say whether researchers in sport and exercise psychology need to increase their sample sizes in future research.

4. The present research

The main goal of the present research was to investigate sample sizes in publications from sport and exercise psychology. We do not aim to introduce the N-Pact Factor as a measure of the quality of a specific journal that might be used like the Impact Factor, similarly to Fraley and Vazire (2014). Given the ongoing discussion on small sample sizes in psychological research, we aimed at assessing the state of the field in sport and exercise psychology. We consider detailed knowledge of the state of the field as a requirement for further steps with the aim of enhancing reproducibility of findings in sport psychology.

Besides our main goal, we investigated into some further questions. First, we were interested in the question of how sample sizes in sport and exercise psychology would fare compared to the sample sizes in social and personality psychology reported in Fraley and Vazire (2014). Second, we hypothesized that sample sizes differ between correlational and experimental studies. This research question can be considered an extension of Fraley and Vazire (2014), who did not investigate into this distinction. Although Fraley and Vazire reported differences between social and personality psychological studies, they did not report whether this difference was confounded with study design. This confound seems likely, as personality psychology studies will use correlational designs more often than social psychological studies. There might also be a general difference between correlational and experimental studies which is independent of studies' theoretical background (i.e., across different theoretical backgrounds, correlational studies have larger sample sizes than experimental studies). Differentiating between correlational and experimental studies is beneficial as it allows a more fine-grained analysis.

5. Method

5.1. Procedure

We closely followed the approach from Fraley and Vazire (2014). We examined studies from four journals in sport and exercise psychology: *International Journal of Sport Psychology* (IJSP), *Journal of Applied Sport Psychology* (JASP), *Journal of Sport and Exercise Psychology* (JSEP), and *Psychology of Sport and Exercise* (PSE). We focused on these four journals as they can be considered the most important journals in sport and exercise psychology and as they have had a Thomson Reuters Impact Factor for the investigation period. We examined all empirical articles published in all four journals starting in 2009 and ending in 2013. We did not examine non-empirical articles such as review articles and comments. After distinguishing between empirical and non-empirical articles, we coded samples sizes of the remaining articles. When an article contained multiple studies that were analysed separately, we coded each study's sample size separately. For every study, we coded in which journal it was published, publication year, sample size, and whether it was an experimental, a quasi-experimental, or a correlational study. We coded a study as experimental, when there was a randomized assignment of participants to factors (between-participants design) or a repeated manipulation (within-participants design). Furthermore, we distinguished between within-participants, mixed (a combination of a within-participants and a between-participants factor), and between-participants designs.

We coded studies as quasi-experimental, when participants' assignment to groups was based on an a-priori existing variable

instead of being randomized (i.e., studies on differences between experts and novices). We coded a study as correlational, when there was no quasi-experimental or randomized assignment of participants to factors. When coding study design, we focused on the operationalization of the main hypotheses. For example, studies, in which hypotheses referred to an experimental design, however contained some additional correlational analyses, were coded as experimental.

We did not analyse meta-analyses, studies based on the analysis of data bases, and qualitative studies such as interview studies or focus groups. Other reasons for not including studies were purely descriptive approaches, studies where we could not identify the relevant sample size, and studies with analyses that do not necessarily rely on large samples (e.g., single case studies). Furthermore, we excluded studies that served the purpose of constructing a questionnaire, although adequate sample sizes are very relevant for questionnaire construction. The reason for this decision is that the rationale for determining adequate sample sizes for questionnaire construction studies does not follow the power logic outlined above (Charter, 1999). Following Fraley and Vazire (2014), we used the initial sample size before cases were excluded. We made exceptions to this rule when the initial sample was only used in order to screen for and identify subgroups that would subsequently be used for the main analyses. In longitudinal studies, we used the sample size at the first measurement time. In dyadic studies, we used the sample size that was the unit of analysis (i.e., in some dyadic studies this is the dyad, in others it is each individual). Coding of sample sizes and design was done by a student research assistant. When the assistant was not sure about a sample size or a design, she asked one of the authors for advice. Importantly, while the research assistant was aware of the overall significance of her task (i.e., the role of sample sizes for quantitative research), she was not aware of the authors' hypotheses regarding the studies' design. One of the authors randomly re-coded 20% of all studies to see whether there would be differences to the first coder which was not the case.

6. Results

6.1. Sample sizes

In total, we analysed 720 studies (see Table 1 for information about how these studies were distributed across the investigation period and the four journals). We did not analyse 339 empirical studies (see Table 2 for information on how these studies were distributed across the investigation period and the four journals). The most frequent reasons for not including a study were either that it aimed at questionnaire construction or that it employed a qualitative approach. We calculated the median sample size across all journals for every year of the investigation period using SPSS 21 (see Table 3). We also calculated the median sample size across all five years of the investigation period, the NF-5.⁴ We calculated the median instead of the mean because the median is less susceptible to outliers or extreme variations of the sample size in general. Furthermore, for every median we calculated the respective

⁴ Here, we diverge from Fraley and Vazire (2014), who calculate the NF-5 as the arithmetic mean of the five annual median sample sizes. When following Fraley and Vazire, one attenuates for annual variations regarding the number of studies published in different years. In other words, every one-year NF is equally weighed, regardless of the number of single studies that contributed to it. To the extent that both sample sizes and numbers of studies vary per year, this makes the NF-5 differ from the "unattenuated" median sample size across all five years. As we were more strongly interested in the latter one, we chose to calculate the median across all sample sizes in all five years.

Table 1
Number of analysed studies per year and journal (final sample).

	2009	2010	2011	2012	2013	Total
IJSP	17	23	27	28	23	118
JASP	19	19	16	14	15	83
JSEP	36	47	38	42	55	218
PSE	54	51	60	69	67	301

Table 2
Number of not-analysed studies per year and journal.

	2009	2010	2011	2012	2013	Total
IJSP	6	4	11	9	12	42
JASP	18	15	14	16	11	74
JSEP	9	13	15	11	15	63
PSE	23	34	26	40	37	160

Table 3
Median sample sizes per year (collapsed over journals).

	2009	2010	2011	2012	2013	5-year
Total	116 (46; 285)	97 (36; 215)	151 (65; 308)	135 (48; 290)	64.5 (33; 226)	114 (42; 257)

Note. 5-year is the median sample size across the whole five year period. Numbers in brackets are interquartile ranges (rounded to the next one-digit number).

Table 4
Median sample sizes per journal and year.

	2009	2010	2011	2012	2013	5-year
IJSP	46 (24; 142)	47 (26; 144)	171 (78; 409)	152 (47; 288)	48 (13; 167)	75.5 (34; 218)
JASP	120 (80; 307)	113 (71; 338)	126 (51; 217)	199.5 (91; 394)	180 (48; 241)	131 (66; 286)
JSEP	93.5 (20; 281)	91 (30; 256)	105.5 (39; 195)	88 (49; 253)	50 (28; 171)	80.5 (32; 224)
PSE	149 (73; 316)	109 (60; 224)	196.5 (81; 411)	133 (38; 317)	126 (34; 277)	138 (55; 298)

Note. 5-year is the median sample size across the whole five year period. Numbers in brackets are interquartile ranges (rounded to the next one-digit number).

interquartile ranges. Interquartile ranges indicate that 25% of samples are smaller than the lower quartile and 25% are larger than the higher quartile.

The NF-5 across all journals is 114. However, there is considerable variation across the five years, with median sample sizes ranging from 64.5 in 2013 to 151 in 2011. Likewise, there is considerable variation within every year, as indicated by the interquartile ranges.⁵ Next, we calculated the median sample size separately for each journal and each year (see Table 4). Again, median sample sizes vary considerably, not only within one journal between years but also between journals. For example, PSE has an NF-5 of 138 whereas IJSP has an NF-5 of 75.5. However, one needs to be extremely careful when comparing median sample sizes between journals, as explained below.

Next we calculated median sample sizes separately for studies employing different designs. The number of studies for this analysis is smaller than the number reported above, as we could not categorize all 720 studies as being either experimental, quasi-experimental, or correlational. Therefore, the sample for the following analyses consists of only 692 studies (see Table 5 for the distribution of studies across years and design categories and Table 6 for the distribution of studies across journals and design categories). We calculated the median sample size across all

journals for every year of the investigation period separately for experimental, quasi-experimental, and correlational studies (see Table 7). A clear trend emerged: Every year, experimental studies had smaller sample sizes than quasi-experimental studies, which in turn had smaller sample sizes than correlational studies. For example, the NF-5 for experiments was 40, whereas it was 91 for quasi-experiments, and 221 for correlational studies. This trend still emerged when looking at each journal separately (see Table 8).

In a next step, we took *only* the studies categorized as experimental and calculated median sample sizes separately for studies employing different experimental designs. This is an important distinction as the same sample size in combination with a given effect size will lead to a higher power in a within-participants design than in a between-participants design. We distinguished between within-participants, mixed, and between-participants designs. The sample for the following analyses consists of only

238 studies (see Table 9 for the distribution of studies across journals and design categories). On average (see Table 10), within-participants studies had smaller sample sizes (NF-5 = 24) than mixed-design studies (NF-5 = 47), which again had smaller sample sizes than between-participants studies (NF-5 = 75). We did not present analyses of median sample sizes separately for journal or year as this would have led to very low numbers per cell in several cases due to the small overall sample size.

The differences between experimental and correlational studies at least partly explain the differences between journals in our sample. Journals with overall larger samples on average published more correlational studies, whereas journals with overall smaller

Table 5
Number of analysed studies per year and design category.

	2009	2010	2011	2012	2013	Total
Exp	32	48	31	56	71	238
Quasi-Exp	25	24	20	26	22	117
Corr	62	62	80	68	65	337
Undefined	7	6	10	3	2	28

Table 6
Number of analysed studies per journal and design category.

	IJSP	JASP	JSEP	PSE
Exp	41	16	100	81
Quasi-Exp	30	15	25	47
Corr	41	42	89	165
Undefined	6	10	4	8

⁵ It is interesting to note that the mean is often a lot higher than the median, sometimes even more than three times higher. This illustrates the observation that sample sizes are heavily skewed, with a few studies having rather high numbers of participants, whereas obviously the minimum number of participants must be cut off at the left end of the distribution.

Table 7

Median sample sizes per year (collapsed over journals), analysed separately for experimental, quasi-experimental and correlational studies.

	2009	2010	2011	2012	2013	5-year
Exp	47 (19; 72)	34 (24; 62)	50 (26; 102)	43 (24; 92)	36 (20; 64)	40 (24; 72)
Quasi-Exp	60 (27; 153)	89.5 (41; 139)	113 (65; 232)	95.5 (52; 278)	70 (38; 213)	91 (45; 206)
Corr	254 (122; 359)	191 (106; 305)	210 (117; 415)	245 (153; 507)	218 (123; 305)	221 (124; 386)

Note. 5-year is the median sample size across the whole five year period. Numbers in brackets are interquartile ranges (rounded to the next one-digit number).

Table 8

Median sample sizes per journal (collapsed over years), analysed separately for experimental and correlational studies.

	IJSP	JASP	JSEP	PSE
Exp	30 (15; 48)	62.5 (32; 103)	40 (24; 74)	44 (25; 101)
Quasi-Exp	80.5 (39; 170)	80 (48; 297)	59 (24; 134)	122 (53; 261)
Corr	225 (155; 453)	202.5 (121; 346)	227 (141; 387)	219 (114; 381)

Note. 5-year is the median sample size across the whole five year period. Numbers in brackets are interquartile ranges (rounded to the next one-digit number).

Table 9

Number of experimental studies per journal and experimental design (within, mixed, between).

	IJSP	JASP	JSEP	PSE	Total
Within	15	1	50	18	84
Mixed	23	12	33	41	109
Between	3	3	17	22	45

Table 10

Median sample sizes of experimental studies (collapsed over journals and years), analyzed separately for design (within, mixed, between).

	5-year
Within	24 (15; 39)
Mixed	47 (30; 85)
Between	75 (44; 124)

Note. 5-year is the median sample size across the whole five year period. Numbers in brackets are interquartile ranges (rounded to the next one-digit number).

samples published more experimental studies (see Table 6). For example, in our sample IJSP had the same number of experimental and correlational studies, whereas PSE had twice as many correlational than experimental studies. Differences between journals are considerably reduced when controlling for study design (see Table 8).

6.2. Power estimations

We further calculated statistical power for a range of typical population effects using G*Power (Faul, Erdfelder, Lang, & Buchner, 2007). As statistical power is a function of three ingredients: alpha, sample size and the population effect size (Cohen, 1977) and alpha is typically set to .05, by convention, we calculated statistical power for a typical study published in the four analysed journals in sport and exercise psychology using the typical sample sizes (i.e., the NF-5) reported in Tables 7 and 10 while assuming a variety of population effect sizes (see Table 11). When calculating power, we differentiated between different study designs. We distinguished between experimental, quasi-experimental, and correlational studies. Within the category of experimental studies, we further distinguished between within-participants, mixed-design, and between-participants studies (see Supplemental Material for more in-depth information on how we calculated power for different designs). We did not only compute the power for the NF-5, but also

for the IQ₂₅ and the IQ₇₅ in order to outline how power varies with varying samples sizes.

Although, there have been repeated calls for reporting effect sizes in the general psychological literature (Cohen, 1994) and in the kinesiology fields (Thomas, Salazar, & Landers, 1991) which have led to an increased reporting in the field of sport and exercise psychology (Andersen, McCullagh, & Wilson, 2007; Ivarsson, Andersen, Johnson, & Lindwall, 2013), little is known (to our knowledge) about typical effects in the field of sport and exercise psychology. Therefore, for the sake of discussion, we will borrow the “typical” effect from the field of personality and social psychology which has been reported to be $r = .20$ ($d = .41$) (Richard, Bond, & Stokes-Zoota, 2003). Although we report power calculations for a range of population effects in Table 11, we consider it informative to discuss a typical effect in psychological science, even if this effect might not be identical in the field of sport and exercise psychology (albeit at present there is no reason to believe that it would substantially differ). This approach enables us to answer the important question of what the power of a typical study (experimental, quasi-experimental, and correlational) in the leading journals of sport and exercise psychology was to detect a typical effect in psychological science.

Table 11 reveals that the power of studies published in the four leading journals in sport and exercise psychology depends on studies' design. The following results always refer to power calculated based on the median sample sizes, meaning that 50% of studies had higher and 50% of studies had lower power (see power calculations for IQ₂₅ and the IQ₇₅). While in general the typical correlational studies in the field of sport and exercise psychology seemed to fare pretty well concerning power, quasi-experimental studies only had a 49% chance of correctly detecting population effects equivalent to a correlation of .20 (Table 11, second column). For experimental studies, it is necessary to further distinguish between designs when calculating power. Studies employing within-participants and between-participants designs again have less than a 50% chance of correctly detecting population effects equivalent to a correlation of .20. Studies employing a mixed design have nearly sufficient power (i.e., .80) for finding effects related to their within-participants factor and interactions. However, they are far less likely to correctly detect effects related to their between-participants factor.

7. Discussion

The goal of the present paper was to introduce the discussion about the so-called crisis of confidence in the mother-discipline of psychological science to the sub-field of sport and exercise

Table 11
Statistical Power to detect various population effect sizes across study categories.

	$r = .1$ ($d = .20$)	$r = .2$ ($d = .41$)	$r = .3$ ($d = .63$)	$r = .4$ ($d = .87$)	$r = .5$ ($d = 1.15$)
Corr	.31 (.20; .50)	.85 (.61; .98)	.99 (.93; .99)	.99 (.99; .99)	.99 (.99; .99)
Quasi-Exp	.16 (.10; .30)	.49 (.27; .83)	.84 (.54; .99)	.98 (.81; .99)	.99 (.96; .99)
Exp-Within	.15 (.11; .23)	.49 (.32; .70)	.84 (.62; .97)	.98 (.88; .99)	.99 (.98; .99)
Exp-Between	.14 (.10; .20)	.42 (.26; .62)	.77 (.53; .93)	.96 (.80; .99)	.99 (.96; .99)
Exp-Mixed					
Within-factor	.27 (.18; .45)	.77 (.56; .96)	.99 (.91; .99)	.99 (.99; .99)	.99 (.99; .99)
Between-factor	.12 (.09; .18)	.34 (.23; .56)	.67 (.47; .90)	.93 (.77; .99)	.99 (.94; .99)
Interaction	.27 (.18; .45)	.77 (.56; .96)	.99 (.91; .99)	.99 (.99; .99)	.99 (.99; .99)

Note. Power values were computed using the NF-5 from [Tables 7 and 10](#) while assuming a range of population effect sizes ($r = .1-.5$). The reported power values assume a two-tailed test with an alpha level of .05. Numbers in brackets denote the power of studies with sample sizes of IQ_{25} and IQ_{75} , respectively.

psychology, while zooming in on the important question of appropriate sample sizes in this field. Most importantly, we aimed at providing insights into the state of the field regarding its sample sizes. While the issue of sample sizes has received some discussion in sport and exercise psychology (e.g., [Andersen & Stoové, 1998](#); [Edvardsson, Ivarsson, & Johnson, 2012](#); [Ivarsson et al., 2013](#)), there has been no systematic investigations on this topic within this field to date.

7.1. Discussion of results

At first sight, our results do not seem to differ much from the results reported by [Fraley and Vazire \(2014\)](#) in the field of personality and social psychology. Fraley and Vazire report their overall NF-5 to be 104, whereas we found ours to be 114. However, the picture looks less favourable when analysing sample sizes separately for different designs. The NF-5 for experimental studies in our sample is 40, with an interquartile range from 24 to 72. This means that 50% of all sport psychological experiments published in sport psychology's most prestigious journals between 2009 and 2013 had sample sizes of 40 participants or less, and 25% of all published experiments even had less than 24 participants. Correlational studies, on the other hand, had larger sample sizes. The NF-5 for correlational studies was 221, which is close to the 250 participants recommended by [Schönbrodt and Perugini \(2013\)](#). On the other hand it is important to note that researchers sometimes divide their overall sample into sub-samples (e.g., when assessing a correlation separately for women and men). In this case, the sample is reduced to 50% of its original size, meaning that the statistical power for detecting an effect is reduced accordingly.

Following from the argument that sample sizes are an important ingredient of quality research as these determine statistical power of a test, we further reported statistical power for a range of typical population effects (see [Table 11](#)). Given the reported sample sizes and corresponding power results for the field of sport and exercise psychology, the implications seem to differ according to which design the study employed. While the correlational studies and the experimental studies employing mixed designs were rather adequately powered, this was not the case for quasi-experimental studies and experimental studies employing within-groups and between-groups designs. Correlational studies probably had larger samples because they are oftentimes more economical as they require little more than distributing pencil-and-paper questionnaires, rendering it easier to sample larger numbers of participants, whereas, experimental studies are often a lot more tedious. However, from a statistical point of view all study designs need to be identically powered to achieve the goals of high quality research: (a) producing precise effect size estimates; (b) producing literature with low false-positive rates; and (c) producing replicable findings ([Fraley & Vazire, 2014](#)).

Once researchers start to emphasize reproducibility and precise

estimations of effects, small samples may be problematic even when they have sufficient power from a NHST perspective. This is because they lead to imprecise estimations of effect sizes (e.g., [Cumming, 2012](#); [Gelman & Carlin, 2014](#); [Schönbrodt & Perugini, 2013](#)). Recommendations for planning sample sizes may differ depending on whether they focus on power for NHST or whether they focus on precision for effect size estimations (e.g., [Cumming, 2014](#); [Schönbrodt & Perugini, 2013](#)). For example, [Schönbrodt and Perugini \(2013, p. 611\)](#) estimate sample sizes necessary for determining the effect size of a correlation with a minimum level of tolerated fluctuations. They conclude that "... for typical research scenarios reasonable trade-offs between accuracy and confidence start to be achieved when n approaches 250".

So far, we have always referred to sample sizes and power while assuming that researchers want to detect a certain effect. However, sometimes researchers want to show that a certain effect does *not* exist. For example, researchers might state that the interpretation of a certain result is only meaningful when there is no between-groups difference related to another factor. Crucially, in order for this reasoning to be valid, the study needs the power to detect the second effect in the first place. According to this rationale, a study should not exclusively be powered to detect the principal effect of interest, but instead it should be planned to have the power to detect the smallest effect that is necessary for its interpretation to be valid. For example, when researchers base the interpretation of a mixed-design study on the between-participants factor not being significant, then the power related to the between-participants factor matters as much as the power related to the within-participants factor.

7.2. Limitations and unintended consequences

The present research has some limitations. One limitation is that we focus on only one characteristic of reproducible research, namely sample sizes. However, large sample sizes are not the only characteristic of research quality. Still, all else equal, larger sample sizes are better. Another limitation is that we did not further distinguish between different research designs. By treating all between-participants studies as having two groups (instead of having more groups) and by assuming that all tests regarding correlational studies were based on their whole sample size (instead of being based on sub-samples), we likely overestimate power for some studies employing between-participants and correlational designs. On the other hand, we may have underestimated the power of studies that either had more than two repeated measurements or for studies that had a higher correlation than .5 between repeated measurements.

Furthermore, we do not know whether sample sizes are sometimes small because researchers assumed the underlying effects to be large. However, as we have argued above, the average effect in psychological research is most likely rather small. Still, it is

important to emphasize that differences between journals may at least partly be caused by different research designs and different assumptions about effect sizes.

Going beyond these limitations of the present research, a more fundamental criticism has been voiced by Fiedler, Kutzner, and Krueger, (2012). They point out that Type 1 errors and Type 2 errors are linked. Reducing the risk of Type 1 errors increases the risk of Type 2 errors. Fiedler and colleagues fear that by one-sidedly focussing on the reduction of Type 1 errors, researchers may underestimate the problems caused by committing more Type 2 errors. Fiedler and colleagues propose some remedies for this dilemma. However, increasing sample sizes does not lead to an inverse relationship between Type 1 and Type 2 error risks: Larger samples protect against both Type 1 and Type 2 errors. Still, one basic problem remains: Assuming that resources for research (e.g., money) remain stable, conducting studies with larger sample sizes means conducting less studies. In other words, large sample studies are more expensive than small sample studies, and this relationship may disproportionately impact different fields of research. For example, increasing sample sizes is less problematic for researchers who conduct questionnaire studies online than for researchers interested in field experiments with particular groups of participants.

Related to the point above is the fact that large sample sizes might compete with other laudable goals in the field of sport and exercise psychology. For example, studies in the area of expertise research might render it difficult to implement large sample sizes due to a small population to draw from. Similarly, multi-method designs, longitudinal designs, or representative designs which are, no doubt, especially needed in the field of sport and exercise psychology might be discouraged by the request for larger sample sizes. Therefore, an unintended consequence of the present discussion would be if these important research streams would become increasingly rare due to the difficulty of sampling larger amounts of participants in these designs.

7.3. Implications for research in sport and exercise psychology

What does the call for adequate sample sizes mean for authors, editors, and reviewers? First, in a straight forward manner, it means that researchers should take the issues of sample sizes seriously and not solely base these on convenience. Instead, researchers should calculate adequate sample sizes a priori based on to-be-expected effects (Cohen, 1977; Faul et al., 2007; Gelman & Carlin, 2014) and based on the degree of uncertainty that they are willing to tolerate (Schönbrodt & Perugini, 2013). We acknowledge that this is no trivial issue, as previously reported sample sizes might themselves have been inflated due to small sample sizes and therefore might not be a reliable estimate of the real world effect. In addition, in most research contexts, researchers do not have a precise expectation concerning the size of the effect (Fraley & Vazire, 2014). In this case, a rule of thumb could be to take the average effect size observed in a field (via meta-analysis) as a helpful way to ground those expectations. In the present manuscript we used the typical effect identified for the field of personality and social psychology (Richard et al., 2003) which was $r = .20$ ($d = .41$). Unfortunately this knowledge is currently not available for the field of sport and exercise psychology and should be determined by future research.

Similarly, editors and reviewers should take the issue of adequate sample sizes seriously. More specifically, they should be alert to large effect size estimates based on small sampled studies. However, this does not necessarily have to be a rejection criterion. One remedy in this case could be to advise to adjust the effect size estimate according to Ivarsson et al. (2013) in order to obtain a

more realistic estimate of the effect in the real world. A course of action that might be more problematic is advising to collect additional data as this can inflate the chance of a Type 1 error/false positive (Sagarin, Ambler, & Lee, 2014). Unfortunately, this course of action seems to be fairly common. Sagarin and colleagues have suggested a feasible remedy in this case by postulating a new statistic $p_{augmented}$, that represents the magnitude of the resulting Type I error inflation and convincingly argue that the “disclosure of post-hoc dataset augmentation via $p_{augmented}$ elevates such augmentation from a questionable research practice to an ethical research decision”. (2014, p. 293).

Throughout this manuscript we have argued that it is important to sample an adequate number of participants for several reasons. However, as mentioned above, this might not be feasible for every research question in sport and exercise psychology. For example in the field of research on expertise, it might sometimes not be possible to have large sample sizes as there is only a very limited population to draw from when one's goal is to investigate the highest level of performers. What could be a possible remedy if a researcher is confronted with this problem? In this case it might be advisable to report adjusted effect size estimates (see Ivarsson et al., 2013) in order to circumvent the problem of inflated effect sizes of small sample studies (see Ivarsson et al., 2013 for further details on adjusting effect sizes).

8. Conclusions

We explicitly admit that research studies we have conducted in the past are not immune to the sample size and power issues raised in this paper. However, with the intention in mind of improving reproducible research in the field of sport and exercise psychology, and especially producing precise effect size estimates and a literature with low false-positive rates, we consider it essential as a field to have an open discussion about this topic in order to ensure adequate sample sizes in addressing the field's research questions. While these types of discussion are currently limited within the field of sport and exercise psychology, the present analysis demonstrates that research in sport and exercise psychology can only benefit from this discussion. In a similar vein, Andersen et al. (2007, p. 671) concluded based on a literature analysis on reporting and interpreting effect sizes in the sport and exercise psychology literature that “the results of our analysis suggest that in current sport and exercise psychology journals, many of the studies published may be somewhat limited in terms of knowledge advancement that is meaningful in the real world”. In line with Fraley and Vazire (2014), it is our hope that this discussion will raise awareness of the issue of adequate sample sizes and researchers from a variety of methodological backgrounds will be encouraged to appropriately scale up their sample sizes in order to produce precise, accurate, and reproducible research.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.psychsport.2015.11.005>.

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